DATA ANALYTICS PORTFOLIO PROJECT

E-Commerce Sales Analysis

Database Design, SQL Analytics & Business Intelligence

5-Table	**1,013**	**Rs. 64.8L**	**26**
Relational Schema	Orders Analyzed	Gross Order Value	SQL Queries Written

Tools: MySQL 8 · SQL (Joins, CTEs, Window Functions) · Python · Pandas · Matplotlib / Seaborn

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Synthetic dataset, generated in-house and kept intentionally clean so the analysis focuses on SQL technique, not data-cleaning.

AGENDA

What This Deck Covers

01 **Project Overview & Schema**

Why this project exists, and how the 5-table database is designed

02 **Key Metrics Dashboard**

Headline revenue, engagement and product numbers at a glance

03 **Five Business Visualizations**

Funnel health, revenue trend, category mix, payments, top customers

04 **SQL Techniques & Problem-Solving**

Joins, subqueries, CTEs, window functions, and the bugs they fixed

05 **Recommendations & Conclusion**

Turning query results into stakeholder-ready action items

SECTION 01

Project Overview & Objective

A self-designed relational database, built and queried end-to-end, not a pre-cleaned CSV.

This project simulates a real online retail business: customers sign up, browse products, place orders, and pay for them. Rather than downloading a ready-made dataset, I designed the schema myself, enforced real referential integrity (foreign keys, CHECK constraints, cascading deletes), and used SQL to answer the kind of ambiguous, multi-table questions a business stakeholder would actually ask.

WHAT THIS PROJECT SET OUT TO DO

- Design a normalized, referentially-sound 5-table schema
- Use joins, subqueries, CTEs & window functions for analysis
- Separate gross order value from net revenue actually collected
- Identify top categories, payment methods & customers
- Translate SQL results into business recommendations

END-TO-END PIPELINE

1. **Schema Design**
   5 tables, foreign keys, CHECK constraints
2. **Synthetic Data**
   300 customers, 1,013 orders, business rules enforced
3. **SQL Analysis**
   Joins → subqueries → CTEs → window functions
4. **Python Visualization**
   Pandas + Matplotlib/Seaborn, 5 charts
5. **Business Recommendations**
   Query results turned into stakeholder-ready insights

SECTION 01

Dataset Overview & Schema Design

A 5-table relational model connected end-to-end by explicit foreign keys.

Table	Key Columns	Rows
customers	customer_id (PK), customer_name, email (unique), city, signup_date	300
products	product_id (PK), product_name, category, price, stock_quantity	32
orders	order_id (PK), customer_id (FK), order_date, order_status	1,013
order_items	order_item_id (PK), order_id (FK), product_id (FK), quantity, price_at_purchase	1,922
payments	payment_id (PK), order_id (FK, unique), payment_method, amount, payment_date	820

TABLE RELATIONSHIP / JOIN PATH

customers → orders → order_items → products,   and   orders → payments (LEFT JOIN: not every order is paid)

Payments only exist for Shipped, Delivered, or Returned orders. *Pending and Cancelled orders deliberately carry no payment row, a modeling decision, not missing data.*

order_status limited to 5 fixed values via CHECK · payments.order_id is UNIQUE (one payment per order) · 6 indexes support every join/filter · price_at_purchase stored on order_items to preserve history

SECTION 02

Key Metrics at a Glance

Headline numbers from the full 1,013-order dataset.

Rs. 64.8L
Gross Order Value

Rs. 52.4L
Net Revenue Collected

80.9%
Payment Collection Rate

Rs. 6,400
Average Order Value

CUSTOMER ENGAGEMENT

Total registered customers	**300**
Placed at least 1 order	**289 (96.3%)**
Repeat customers (2+ orders)	**257 of 289 (88.9%)**
One-time customers	**32 of 289 (11.1%)**
Average units purchased per order	**3.81**

TOP-SELLING PRODUCTS (BY REVENUE)

Product	Units	Revenue (Rs.)
Portable SSD 1TB	87	5,65,413
Mixer Grinder	119	4,16,381
Wireless Earbuds	136	4,07,864
Smartwatch	107	3,74,393
Running Shoes	118	3,30,282

Order Funnel: Status Distribution

All 1,013 orders broken down by current status - a quick read on funnel health.

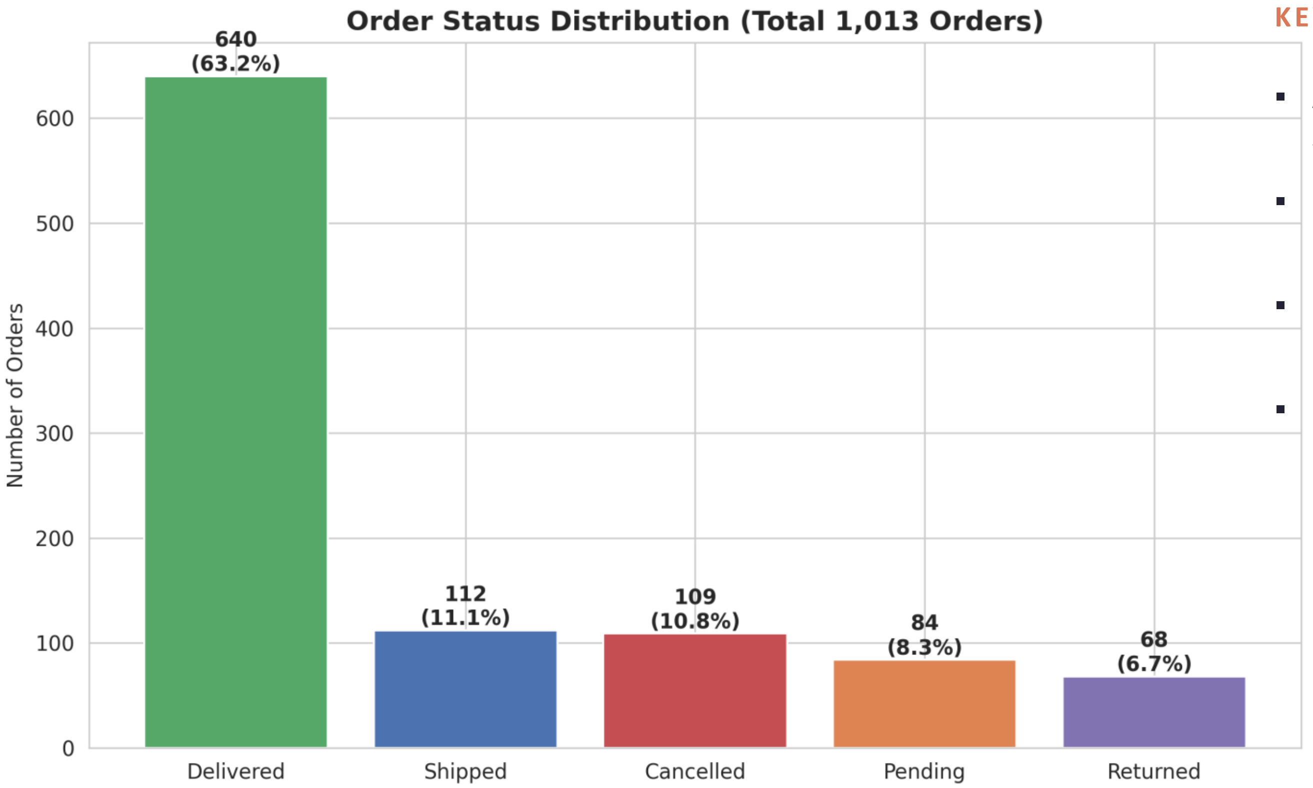

Fig. Distribution of all 1,013 orders by current status.

KEY TAKEAWAYS

- About 3 in 4 orders reach Shipped or Delivered - the core pipeline is working.
- Cancellations (10.8%) outnumber Returns (6.7%) - more value slips away before payment than after delivery.
- Cancelled + Returned together sit at 17.5% - that's the number I'd keep an eye on.
- Pending sits at 8.3% - not a problem on its own, but worth watching if it grows relative to Delivered.

Monthly Revenue Trend

Gross order value vs. what actually gets collected as revenue, month by month.

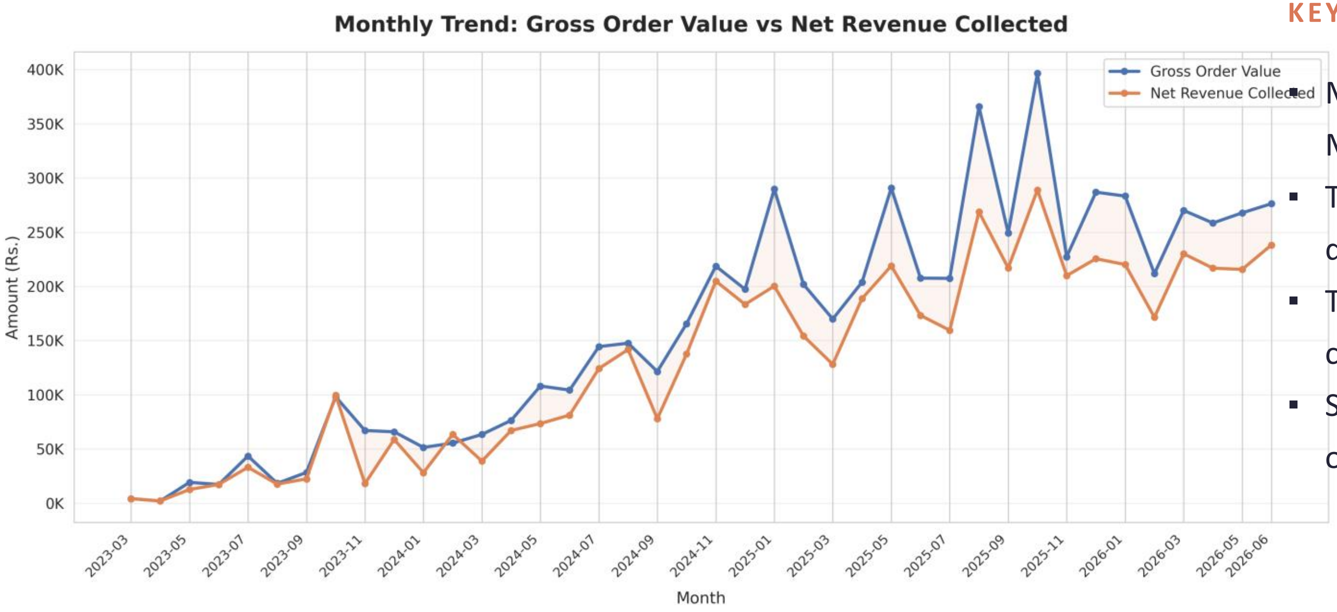

Fig. Monthly gross order value vs. net revenue collected, Mar 2023 – Jun 2026.

KEY TAKEAWAYS

- Monthly order volume climbed roughly 63x, from under Rs. 10K in Mar 2023 to over Rs. 2.65L in the latest quarter.
- That growth holds up across almost 3 years - this looks like real demand, not a one-off spike.
- The gap between gross and net widens in busier months - collection cycles are worth watching as volume grows.
- Skip the payments join and a dashboard built off orders alone overstates real cash collected by about 19%.

Category-Wise Revenue

Gross order value by product category - and which ones actually carry the business.

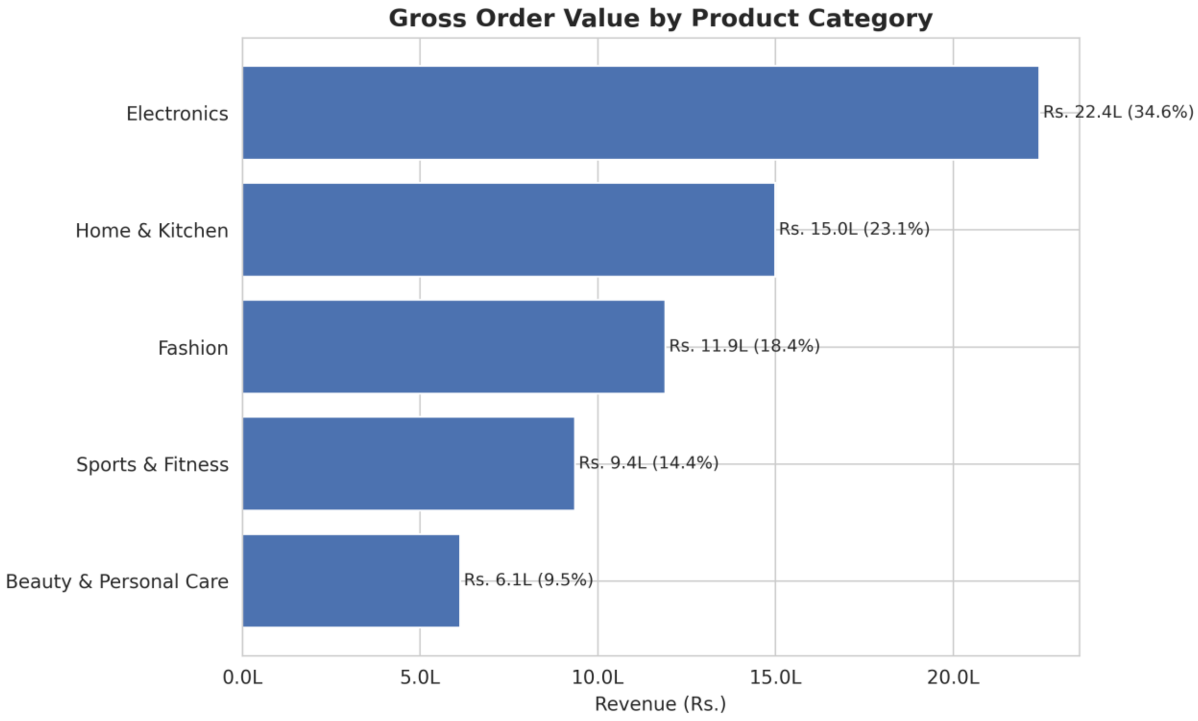

Fig. Gross order value by product category.

KEY TAKEAWAYS

- Electronics leads by a wide margin - 34.6% of gross order value from just 8 of 32 products.
- Electronics and Home & Kitchen together make up 57.7% of order value - worth using as a concentration benchmark.
- Beauty & Personal Care sits at just 9.5%, despite a healthy price point - the clearest candidate for catalog expansion.

Payment Method Usage

820 completed payments, broken down by method.

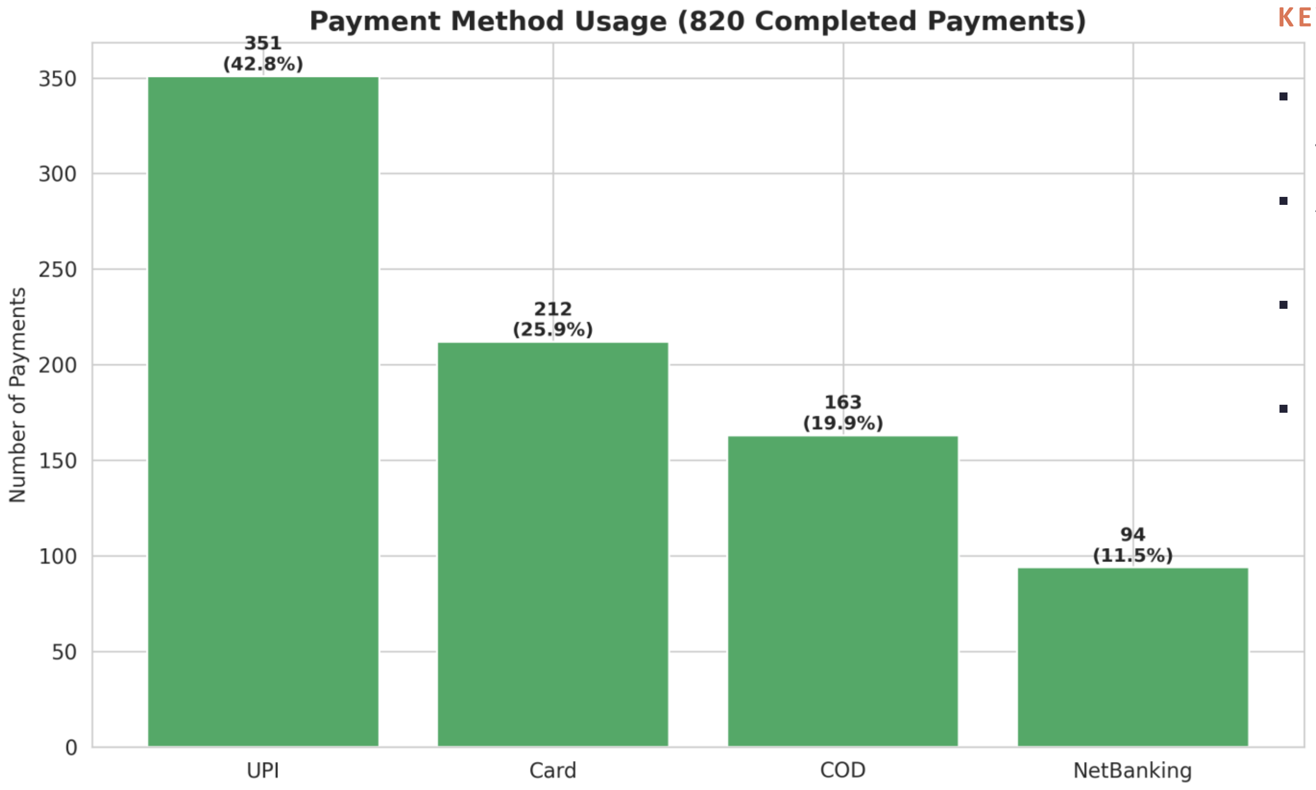

Fig. Count and share of completed payments by method.

KEY TAKEAWAYS

- UPI leads clearly with 351 payments - 42.8% of volume, in line with the wider Indian e-commerce market.
- Any downtime or failed-transaction handling on UPI would hurt conversion more than a problem on any other rail.
- UPI, Card, and NetBanking together cover over 80% of volume - most revenue settles instantly instead of sitting in COD delays.
- COD's 19.9% share actually understates its real cost, since a payment only shows up once the order ships.

Top 10 Customers by Spend

How dependent is the business on its top spenders? Answered with a full 5-table join.

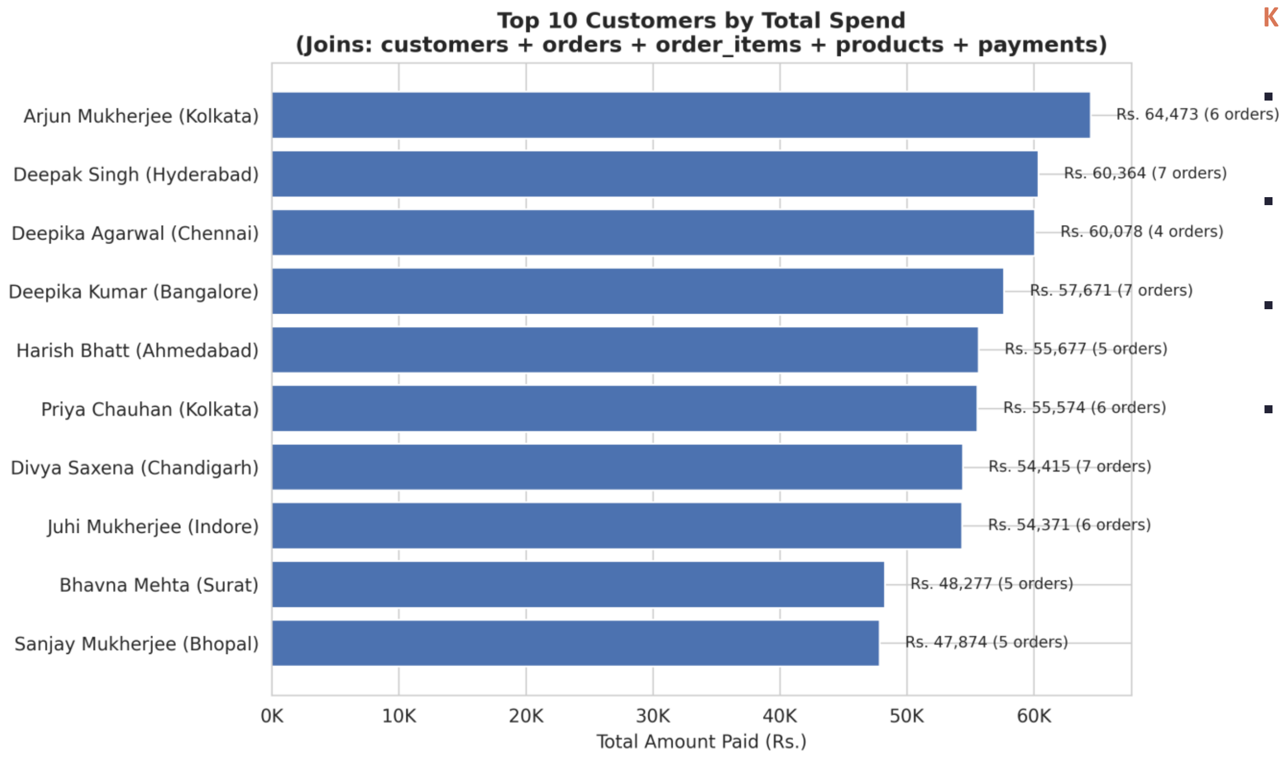

Fig. Top 10 customers by total amount paid.

KEY TAKEAWAYS

- Top spender: Arjun Mukherjee (Kolkata), Rs. 64,473 across 6 orders. #10: Sanjay Mukherjee, Rs. 47,874 across 5 orders.
- No single customer dominates - the top 10 combined are a small, healthy slice of the Rs. 52.4L total net revenue.
- What separates them is order frequency (5-7 orders each), not one big-ticket purchase.
- 109 of 289 active customers - about 1 in 3 - spend above average, a solid-sized segment for a loyalty program.

SECTION 04

SQL Techniques & Problem-Solving

The techniques you'd expect in a data analyst role, and the bugs each one was built to catch.

Technique	Applied To
Multi-table JOIN (INNER + LEFT)	5-table chain: customers → orders → order_items → products → payments
Subquery in FROM / HAVING	Order-level aggregation before joining payments; above-average spenders
CTE (WITH clause)	Monthly gross order value as a reusable result set
Window Fn: SUM() OVER / RANK() / LAG()	Running totals, spend leaderboard, month-over-month growth %
NULLIF() + EXPLAIN	Safe division for growth %; verifying index usage on key filters

REAL PROBLEMS THIS CAUGHT

Fan-out on JOIN: Joining order_items straight to payments would've inflated totals, so I aggregated to order-level first.

Silent row loss: An INNER JOIN would silently drop unpaid Pending/Cancelled orders, so I used LEFT JOIN wherever completeness mattered.

Fragile GROUP BY: Grouping by customer_name alone risked merging two people with the same name, so I added customer_id too.

Divide-by-zero: Growth % is undefined in month one, so I wrapped the denominator in NULLIF(prev_month_value, 0).

SECTION 05

Real-World Problems & Data-Backed Solutions

Three findings that translate directly into business action.

01

Revenue Reports That Overstate Reality

The numbers: Gross order value is Rs. 64.8L, but net revenue actually collected is Rs. 52.4L, a 19.2% gap from orders that never generate a payment. **What I'd do:** Join orders to payments with LEFT JOIN and report two clearly labeled metrics: gross_order_value and net_collected_revenue. Never one blended number.

02

Cancellations Outpacing Returns

Here's what stood out: 109 orders (10.8%) were Cancelled vs. 68 (6.7%) Returned, more value is lost before payment than after delivery. **The fix:** Prioritize checkout-funnel fixes (payment failure handling, stock checks) over post-delivery returns handling, a cheaper, higher-leverage fix.

03

Category Revenue Concentration

What the data shows: Electronics + Home & Kitchen = 57.7% of gross order value; Beauty & Personal Care contributes under 10%. **My take:** Track category mix as a recurring KPI, not a one-time observation, to flag concentration risk early.

STRATEGIC RECOMMENDATIONS

Finance & Reporting

Standardize on net_collected_revenue as the headline figure; keep gross_order_value as a secondary funnel-health metric.

Operations

Root-cause the 109 Cancelled orders before investing further in post-delivery Returns handling, the larger, cheaper lever.

Merchandising

Evaluate a targeted expansion of the Beauty & Personal Care catalog, currently the smallest contributor at 9.5%.

Payments & CRM

Monitor UPI checkout reliability (42.8% of volume) and build a loyalty tier around the 109 above-average-spend customers.

CONCLUSION

This 5-table, 1,013-order database goes all the way from schema design to business-facing analysis - the picture that comes out is a fundamentally healthy business, with two things worth acting on: cancellations meaningfully outpace returns, and revenue leans heavily on 2 of 5 categories.

63.2%
Order-Delivery Rate

80.9%
Payment Collection Rate

Zero
Single-Customer Risk

LET'S CONNECT

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